

DHT11 Climate Sensor

Separate transport, validity, freshness, and physical accuracy

The RGB cadence reports software state; TP13 reveals the wire.

Lesson	013 — component	Time	120–180 minutes
Prerequisites	Lessons 001–012	Board	Arduino Mega 2560
Evidence	D5–D7 status indicator and TP13	Status	host verified; physical acceptance open
Safety	E1; USB-powered low-current circuit only		

Safety checkpoint. Disconnect USB before wiring. Identify the exact bare sensor or module before adding a DATA pull-up: many modules already contain one. Never connect the DHT11 NC pin. Stop for heat, odor, resets, unstable power, a DATA voltage outside the supply rails, or disagreement between the table and the identified part. Remove USB power to stop the experiment.

1 Mission

Build a scheduled climate observation path:

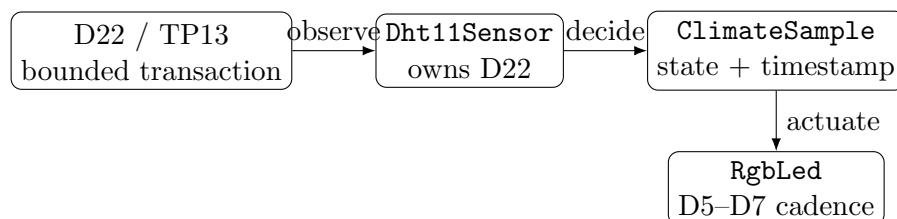
DHT11 at D22 → `Dht11Sensor` → `ClimateSample` → D5–D7 status evidence.

The lesson deliberately does not add a display. Numeric presentation belongs to lesson 014. Here the learner asks four narrower questions:

1. Did the software acquire its pin resource?
2. Did a bounded wire transaction complete?
3. Did checksum and range validation accept the decoded sample?
4. Is the last accepted sample still fresh under the lesson policy?

Those claims are independent. A pulse train at TP13 does not establish climate accuracy. A green indicator means the decoded sample is fresh and valid, not that the sensor has been calibrated against a reference instrument.

2 Composition and lifecycle



Construction is inert. `initialize()` validates and claims D22, leaves the pulled-up DATA wire released as an input, and sets the sample to `Unavailable`. Repeated initialization is harmless. A failed RGB acquisition shuts the sensor down before setup returns.

`update(now)` schedules the next eligible transaction from caller supplied time. It never uses an unbounded wait. An early call is a successful no-op. `sample(now, freshFor)` reads state without touching hardware.

`shutdown()` releases DATA to high impedance, clears the snapshot, releases the claim, and is idempotent. The destructor repeats that cleanup safely.

```
void loop ()
{
  const adk::TimePoint    now      (millis ());
  const adk::Status       status   = observeClimate    (now);
  const adk::ClimateSample observation = decideClimateState (now);

  showClimateState (observation, now);
}
```

The canonical sketch additionally checks each status and halts with outputs off if the evidence circuit itself fails.

3 Parts and orientation

Quantity	Part	Requirement
1	Mega 2560	USB powered
1	DHT11	exact bare part or documented module identified
1	DATA pull-up	value from the identified part; omit if module supplies it
1	common-cathode RGB LED	each die identified before wiring
3	330 Ω resistors	one for each RGB channel
1	0.1 μ F capacitor	across sensor VDD/GND when part guidance calls for it
1	logic analyzer or oscilloscope	optional TP13 timing evidence
1	multimeter	continuity, rail, and released-level checks

4 Literal wiring

Function	Mega connection	Literal path
DHT11 supply	5 V and GND	VDD to 5 V; GND to GND; never use Vin
DATA	D22	D22 to identified DATA pin; label the sensor-side node TP13
NC	none	leave electrically unconnected
pull-up	DATA to 5 V	fit only the value required by the identified part/module
red evidence	D5	D5 through 330 Ω to red anode
green evidence	D6	D6 through 330 Ω to green anode
blue evidence	D7	D7 through 330 Ω to blue anode
RGB return	GND	common cathode to GND

Check resistance and continuity with USB removed. Verify that the pull-up does not become an unintended parallel resistor on a module. Place the meter or analyzer clip at TP13 while unpowered; inspect the clip spacing before applying USB power.

5 Wire transaction

The adapter drives DATA low only for the request, then releases it:

output low $\rightarrow$ input/high impedance $\rightarrow$ sensor response $\rightarrow$ 40 data bits.

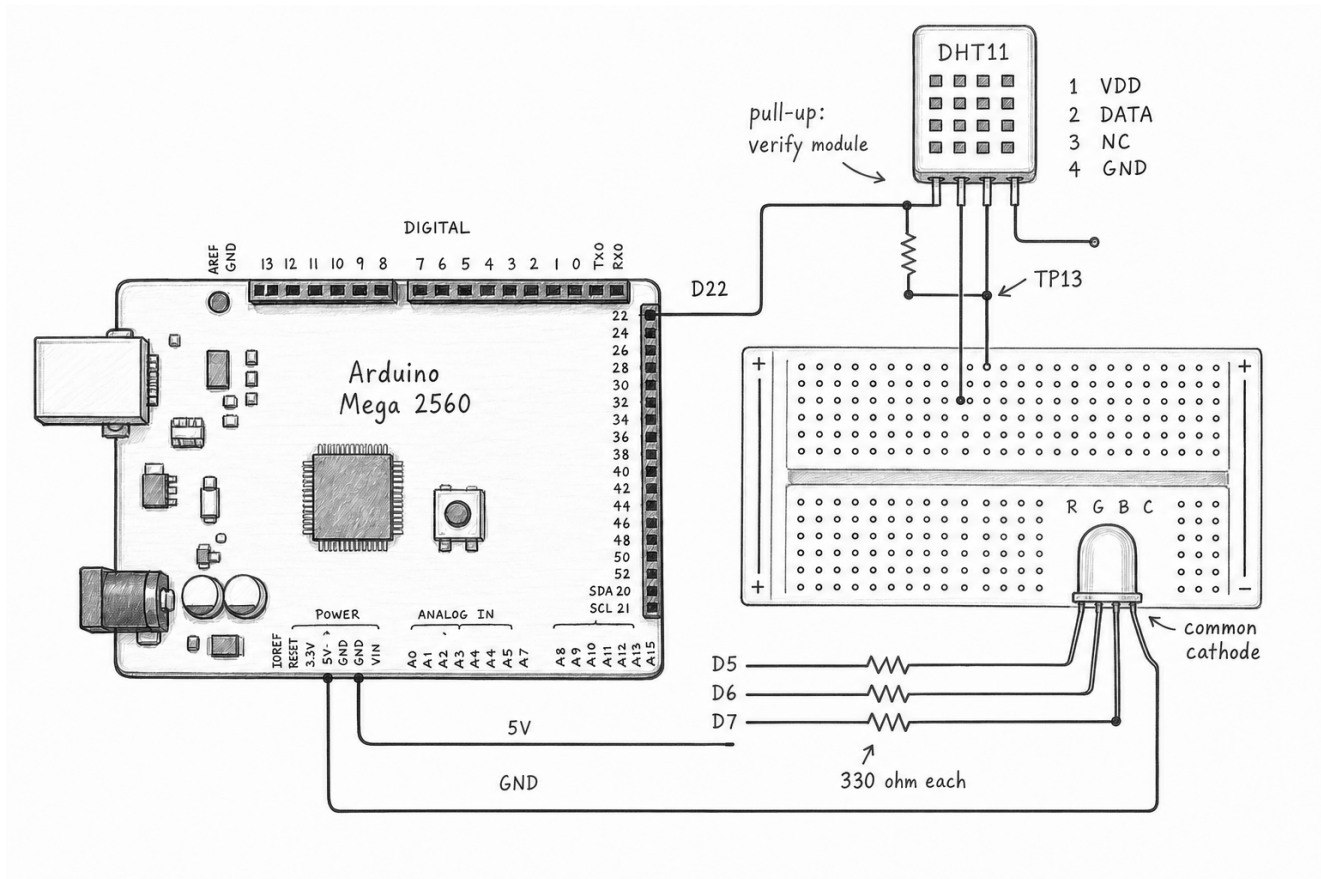
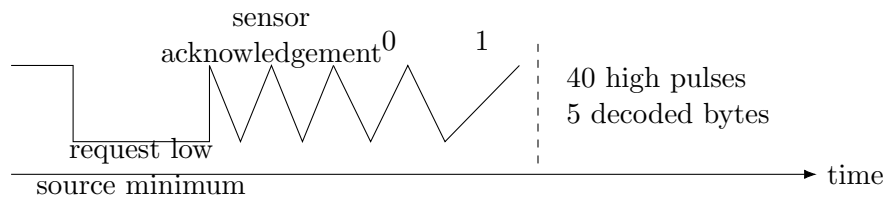


Figure 1: Pencil orientation of the Mega, DHT11, DATA pull-up, RGB evidence, and TP13. This is not an authoritative schematic. Pin placement and breadboard connectivity are simplified; use the literal table and the exact part documentation. Alternative text: Mega D22 reaches DHT11 DATA and TP13; D5, D6, and D7 reach three resistor-limited RGB channels; all parts share 5 V and GND while NC remains unconnected.



The drawing shows shape, not measured scale. Record the selected manual's request, acknowledgement, zero, and one windows before comparing a capture. Every edge wait has a finite deadline. The five bytes carry humidity, temperature, and a checksum under the exact selected DHT11 revision. Do not infer DHT22 compatibility.

6 Visible state vocabulary

Color is supporting evidence; pulse count and cadence carry the distinction in black-and-white observations.

Sample state	D5–D7 indication	Interpretation
Unavailable	one slow blue pulse	resource owned; first frame pending
Valid	steady green	checksum, range, and freshness accepted
TransportTimeout	two red pulses	bounded transaction did not complete
ChecksumFailure	three red pulses	40 bits arrived but integrity failed
range failure	three red pulses, then amber	decoded value outside limits
Stale	steady amber	earlier valid sample exceeded freshness policy
setup/evidence failure	indicator off	circuit halted; inspect acquisition separately

TP13 is the primary transport evidence. Predict that released DATA rests high, observe the request and response, then interpret only whether transport occurred. A meter can verify the resting level, but a logic analyzer or oscilloscope is required to inspect pulse timing.

7 CLI workflow

```
make host-test-climate-sensor
make host-test-dht11-sensor
make sanitize
make arduino-Lesson013Dht11Climate
make size-check
make lesson-013
make lessons-check

make upload-Lesson013Dht11Climate PORT=/dev/ttyACM0
```

Use the repository’s actual focused target names if integration reports a different spelling. Serial is neither required nor used by the canonical example. Upload and evidence collection remain CLI-driven.

8 Predict, observe, interpret

8.1 Acquisition and first frame

Predict. Immediately after successful acquisition, one slow pulse repeats while the adapter honors power-up and acquisition intervals.

Observe. Status cadence: _____. TP13 resting voltage: _____ V relative to GND. Time to first green: _____ s.

Interpret. A waiting pulse proves the software objects initialized. A high TP13 resting voltage is separate electrical evidence that DATA is released. Steady green means one sample passed protocol, checksum, range, and freshness checks.

8.2 Reversible transport fault

Remove USB power, disconnect only DATA, inspect the circuit, then restore USB. Do not short DATA to either rail.

Predict. The bounded transaction times out and two red pulses repeat.

Observe. Pulse count: _____. Time from request to fault: _____. TP13 waveform present? ☐ yes
☐ no.

Interpret. A timeout distinguishes missing wire response from checksum failure. It does not locate the cause; wiring, part orientation, pull-up, or sensor health may be responsible.

8.3 Recovery and freshness

Remove USB, restore DATA, inspect, then apply USB. A later valid frame should replace the transport fault normally. To study staleness deterministically, use the host recorded-sensor test; do not invent an electrical fault that pretends time advanced.

Observe. Recovery cadence: _____. Host stale boundary: _____ ms. State at the exact boundary: _____.

9 Diagnosis

Evidence	Likely boundary	Next safe check
indicator always off	resource or RGB acquisition	remove USB; inspect pins, resistors, and resource conflicts
waiting never ends	power-up interval or no eligible response	inspect TP13 with a bounded capture and confirm the selected part
two red pulses	transport timeout	remove USB; inspect DATA continuity, orientation, supply, and pull-up decision
three red pulses	checksum failure	shorten wiring, inspect supply integrity, and compare measured timing to the selected manual
three red then amber	decoded range failure	verify part revision and byte interpretation; do not present numeric fields as valid
steady amber	freshness policy	inspect scheduling and last valid timestamp
green but implausible climate	physical accuracy, not transport	compare against a suitable reference; record uncertainty

10 Exercises

1. Explain why `sample(now, freshFor)` must not trigger hardware.
2. A frame has valid timing but the checksum differs. Which state and cadence should appear?

3. A valid sample becomes old. Why must it become `Stale` rather than `TransportTimeout`?

4. Describe why driving a pulled-up open-style DATA line high is less safe than releasing it to input.

5. Choose a freshness interval for a room display and justify it without claiming a faster DHT11 acquisition rate.

11 Bench acceptance record

Status before this card is signed: host verified; hardware acceptance open. Compilation, tests, and this document do not supply physical evidence.

Date / reviewer	_____
Board / USB supply	_____
Sensor or module identity	_____
Primary manual and revision	_____
Pull-up fitted/on module	_____
Measured 5 V rail	_____ V
RGB resistor values	red _____ Ω , green _____ Ω , blue _____ Ω
TP13 released level	_____ V relative to GND
Request low width	minimum _____, maximum _____
Acknowledgement widths	low _____, high _____
Observed zero/one highs	zero _____, one _____
Reset / shutdown observation	_____
Disconnected-DATA result	_____
Recovery result	_____
Power-removed continuity	_____
Deviations / stop events	_____

Acceptance claim. Resource evidence, released-wire evidence, successful transport, decoded validity, freshness, and physical accuracy are separate observations. Sign only the claims actually measured.